

Home hub — routine cheat sheet

What to do, where, and when, to keep sync, the phone pages and the backups working. Full details: [~/sys_org/home_hub/ \(README → runbook, helpers, recovery\)](#). Home laptop = Arch, user archx, tailnet name home.tail53b947.ts.net, SSH port 2255.

1 · Every study or work session

Where / when	Command	What it does
BOTH sit down	sync-all	Pulls study system, lectures, org-roam, home_hub, Timewarrior; files the phone inbox; task sync. Asks the GitHub key passphrase (no agent). Read the last line: All synced or Needs attention: <repo> .
BOTH get up	sync-all	Commits and pushes what you changed. The one rule: never edit the same file on both laptops between two syncs.
BOTH dry run	sync-all -n	Shows what would be synced; touches nothing.
WORK shell at home	ssh -t home tmux new -A -s main	Reattaches the same session after a drop. Plain ssh home also works (key id_ed25519_home).
WORK notebooks, webUIs	http://home.tail53b947.ts.net:8881	Containers published on the tailnet address (docker run -p \$(tailscale ip -4):8881:8888 ...). Ports: 888x Jupyter · 42xx Quarto · 80xx web.
WORK 127.0.0.1-only port	fwd 8881 4200 unfwd 4200 ssh -O exit home	On-the-fly tunnels over one shared SSH connection (the fwd function in ~/.bashrc, ControlMaster on Host home).
HOME heavy work	nvim file.qmd # in tmux quarto preview ... --port 4200	Code, kernel, data and renderer on the home laptop; only keystrokes and the page cross the network.

2 · Phone (Android) and clusters

Where / when	Do	Note
PHONE always	Tailscale app connected	Nothing is reachable without it; it can stay on all day.
PHONE read	https://home.tail53b947.ts.net/ https://home.tail53b947.ts.net:8443/notes/week.html	Lectures (progress saved, committed within 10 min) · study site. Pin both: : → Add to Home screen.
PHONE note / log	logs/inbox.md ← one line time 06:00-06:50 math ... next math: ... win: ...	Filed by the next sync-all on a laptop (learn inbox).
PHONE shell	ssh home # Termux	Termux config uses the 100.x address (tailscale ip -4 at home), port 2255.
CLUSTER code, notes	git clone / git pull (GitHub)	Clusters cannot join the tailnet. Log time there with an inbox line, commit, push.
CLUSTER data → drive	sh ~/sys_org/home_hub/bin/backup-clusters.sh	From the home laptop if the cluster lets it in; else from the work laptop (→ ~/cluster-mirror, carried by its backup). List: ~/.config/home-server/clusters.

3 · What runs by itself (nothing to do — just know it is there)

	Job	When	What
HOME	home-refresh	every 10 min	Pull + rebuild the study site in ~/srv, commit phone reading progress, sync Timewarrior, task sync. Uses the passphrase-less deploy keys.
BOTH	task-keep	home 03:00 work 01:00	JSON export of all tasks → ~/backups/tasks/ (90 days); at home also a copy of the sync server DB (14 days).
HOME	backup-system (root)	daily 03:30	Whole system + /mnt/StorageHDD + manifest → Toshiba drive; refreshes backups/RECOVERY/.
WORK	backup	01:30	/home/hlima → drive over Tailscale (backup key); skips quietly when home is unreachable, catches up later.
HOME	backup-prune (root)	Sun 05:00	Thin snapshots (7 daily · 8 weekly · 24 monthly · 10 yearly), free space, restic check — every repository.
HOME	backup-check (root)	1st, 06:00	Reads back 10 % of every repository's data.
HOME	lectures-web · study-web · tasksync	always	On 127.0.0.1, published to the tailnet by tailscale serve on 443 · 8443 · 10000.

4 · Weekly — at the Saturday review (5 minutes)

HOME	<code>sudo backup-system snapshots tail -3</code>	A snapshot from last night?
WORK	<code>sh ~/sys_org/home_hub/bin/backup.sh snapshots tail -3</code>	A snapshot from the last day or two?
BOTH	<code>systemctl --failed; systemctl --user --failed</code>	Must list 0 units.
HOME	<code>journalctl -u backup-prune -n 15 --no-pager</code>	Ends with “no errors were found”.
HOME	<code>journalctl --user -u home-refresh -n 10 --no-pager</code>	No “pull failed” / “build failed”.
BOTH	<code>tailscale status</code>	Both laptops and the phone listed; home “active”.

5 · Monthly — at the monthly review (15 minutes)

BOTH	Restore one file and open it — the only real proof	<code>sudo RESTIC_PASSWORD_FILE=/etc/restic/password restic -r \</code> <code>/run/media/archx/Toshiba_4T/backups/restic/\$(hostname -s) \</code> <code>restore latest --include <file> --target /tmp/r # HOME</code> <code>sh ../backup.sh restore latest <file> /tmp/r # WORK</code>
HOME	Disk health (old laptop, old disks)	<code>cat /var/backups/system-manifest/smart.txt</code> <code>journalctl -u backup-system grep -i smart</code>
HOME	Drive space	<code>df -h /run/media/archx/Toshiba_4T</code>
HOME	Monthly read-back passed	<code>journalctl -u backup-check -n 10 --no-pager</code>
BOTH	Task exports exist from today	<code>sh ~/sys_org/home_hub/bin/task-keep.sh --list tail -3</code>
HOME	Serving clone clean (never edit it)	<code>git -C ~/srv/study-system status -s</code>
HOME	Updates , then check everything came back	<code>sudo pacman -Syu && sudo reboot</code> # afterwards: phone pages load, task sync works
WORK	Updates	<code>sudo apt update && sudo apt upgrade</code>

6 · Quarterly and yearly

Quarterly	Second copy away from the house (fire, theft, surge take laptop + drive together):	<code>restic -r /run/media/archx/OFFSITE/restic-home copy \</code> <code>--from-repo</code> <code>/run/media/archx/Toshiba_4T/backups/restic/<host> latest</code>
Quarterly	Cluster data to the drive	<code>sh ~/sys_org/home_hub/bin/backup-clusters.sh</code>
Yearly	Rehearse a rebuild in a VM (recovery.org, scenario 3 or 4). Note every surprise in the file.	<code>virt-manager # or qemu, with the real drive attached</code>
Yearly	Lectures repository size (each library rebuild adds history)	<code>gh repo view h-lima/lectures --json diskUsage</code>
Yearly	Arch USB stick still boots; password manager holds every secret (box below).	

7 · Before anything risky, and after editing a helper

Before upgrading Taskwarrior, touching ~/.task or ~/.timewarrior, merging, or any change you are unsure about	sh ~/sys_org/home_hub/bin/safety-snapshot.sh	Must end with SAFETY SNAPSHOT COMPLETE . Roll back tasks: mv ~/.task ~/.task.broken && tar xzf S/dot-task.tar.gz -C ~
After editing bin/backup-system.sh in home_hub	sudo sh ~/sys_org/home_hub/bin/backup-system.sh install	Root runs only the installed copy in /usr/local/sbin.
After changing config.sh or the services	sh ~/sys_org/home_hub/bin/install-home.sh	Safe to run again.
New excludes on the work laptop	restic backup ~ --exclude-file ~/.config/restic/excludes \ --one-file-system --dry-run -v tail -5	See size and file count before a real run.

8 · When something is off

Phone page does not load	tailscale serve status systemctl --user status lectures-web study-web	App connected? Services active?
Progress does not save	grep LIBRARY_HOSTS ~/.config/systemd/user/lectures-web.service	Must be the exact tailnet name.
Study site out of date / no PDF	less ~/srv/build-site.log	reportlab in ~/srv/study-system/.venv?
task sync fails	ssh home systemctl --user status tasksync task show sync	Settings identical on both laptops. Worse: recovery.org scenario 7.
Backup “not reachable”	ssh home true sftp home-backup # then: bye ssh home mountpoint /run/media/archx/Toshiba_4T	Tailscale up? Drive mounted?
Needs attention: <repo>	cd <repo> && git status	Resolve, git rebase --continue, run sync-all again.
Pull stops in ~/.timewarrior	python3 ~/sys_org/home_hub/bin/timew-merge.py install	Driver missing in that clone; then continue the rebase.
SSH asks for the wrong key	ssh -v home true 2>&1 grep -iE 'offering accepted'	IdentitiesOnly yes in the Host entry.
Stale restic lock	restic unlock	Removes only locks of crashed runs.

9 · Keys, secrets, and where to look

Key	Opens	Lost a device? Revoke its keys:
~/.ssh/id_ed25519	GitHub (work laptop)	GitHub → Settings → SSH keys (and repo → Deploy keys)
~/.ssh/id_ed25519_home	home laptop login	home laptop: delete its line in ~/.ssh/authorized_keys (each line has a comment)
~/.ssh/id_ed25519_backup	backups only: sftp, tailnet only	In the password manager — never only on a laptop:
~/.ssh/deploy_* (home)	home-refresh: one repo each	restic password · Taskwarrior sync. encryption_secret · key passphrases
Termux key (phone)	home laptop login	If a machine dies: ~/sys_org/home_hub/recovery.org — also on the drive in backups/RECOVERY/, readable without restic.